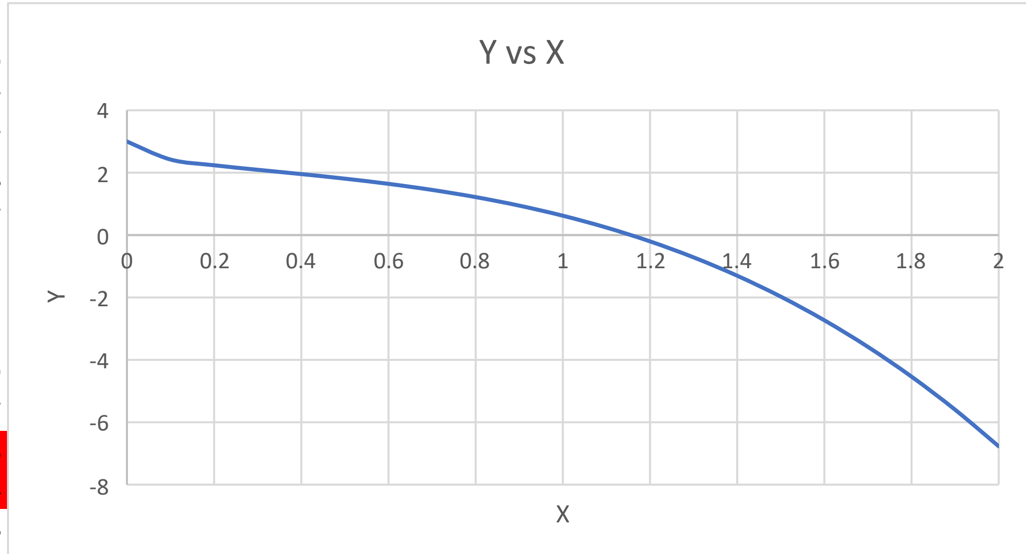
f= 3cos(x^1/5)-x^3

X	Y
0	3
0.1	2.421389
0.2	2.237937
0.3	2.093037
0.4	1.954968
0.5	1.808217
0.6	1.642053
0.7	1.447735
0.8	1.217512
0.9	0.944189
1	0.620907
1.1	0.241028
1.2	-0.20194
1.3	-0.71438
1.4	-1.3026
1.5	-1.97286
1.6	-2.73136
1.7	-3.58429
1.8	-4.53778
1.9	-5.59797
2	-6.77097



Initial x 1 Rearranged Equation: (3*cos(x^(1/5))^1/3

Fixed Point Iteration Method

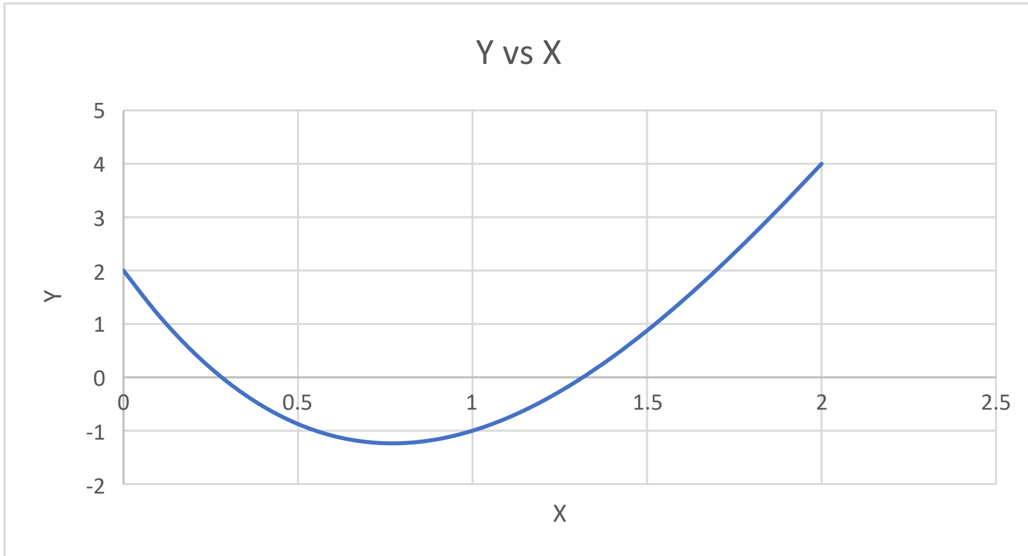
Step	X	xi+1	delta
1	1	1.174679	0.174679
2	1.174679	1.154162	0.020517
3	1.154162	1.156497	0.002335
4	1.156497	1.15623	0.000267
5	1.15623	1.15626	3.04E-05
6	1.15626	1.156257	3.47E-06
7	1.156257	1.156257	3.97E-07

Solver

x	y calc	offset
1.156257	-1.6E-07	1.55E-07

f(x)	2-9x+7x^2-x^3

X	Y
0	2
0.1	1.169
0.2	0.472
0.3	-0.097
0.4	-0.544
0.5	-0.875
0.6	-1.096
0.7	-1.213
0.8	-1.232
0.9	-1.159
1	-1
1.1	-0.761
1.2	-0.448
1.3	-0.067
1.4	0.376
1.5	0.875
1.6	1.424
1.7	2.017
1.8	2.648
1.9	3.311
2	4



f(x)	2-9x+7x^2-x^3
f'(x)	-9+14*x-3x^2

Newton Raphson 1

x	f(x)	f'(x)	x+1	offset
1	-1	2	1.5	0.5
1.5	0.875	5.25	1.333333	0.166667
1.333333	0.074074	4.333333	1.316239	0.017094
1.316239	0.000882	4.229893	1.316031	0.000208
1.316031	1.33E-07	4.228621	1.316031	3.13E-08
1.316031	0	4.22862	1.316031	0

Newton Raphson 2

x	f(x)	f'(x)	x+1	offset
0.5	-0.875	-2.75	0.181818	0.318182
0.181818	0.589031	-6.55372	0.271696	0.089877
0.271696	0.051413	-5.41772	0.281185	0.00949
0.281185	0.000556	-5.3006	0.28129	0.000105
0.28129	6.78E-08	-5.29931	0.28129	1.28E-08
0.28129	7.46E-16	-5.29931	0.28129	1.67E-16
0.28129	3.82E-17	-5.29931	0.28129	0

Solver

x	y calc	Offset
0.28129	3.24E-07	3.24E-07
1.316031	3.3E-07	3.3E-07

Constants

R	T	a	b	Tc	Pc	n
0.00008206	373	1.72E-05	2.21E-05	154.4	49.7	2

P1	P2
0.009869	9.869

F(v)

$$\frac{RT}{\frac{V}{n} - b} - \frac{a}{\sqrt{T} \cdot \frac{V}{n} \left(\frac{V}{n} + b \right)} - P$$

F'(V)

$$\frac{an}{\sqrt{T}\, V^2 \cdot \left(\frac{V}{n} + b \right)} + \frac{a}{\sqrt{T}\, V \cdot \left(\frac{V}{n} + b \right)^2} - \frac{RT}{n \cdot \left(\frac{V}{n} - b \right)^2}$$

Vi+1 = Vi + F(v)/F'(V)

NR Method

P1

Step	V	F(v)	f'(v)	vi+1	delta
1	1	0.051347	-0.0612	1.838944	0.838944
2	1.83894365	0.02342	-0.0181	3.132838	1.293894
	3.13283761	0.009671	-0.00624	4.6835	1.550662
	4.68349989	0.003202	-0.00279	5.830779	1.147279
	5.83077883	0.00063	-0.0018	6.180605	0.349826
	6.18060469	3.56E-05	-0.0016	6.202841	0.022236
	6.20284088	1.27E-07	-0.00159	6.202921	7.96E-05
	6.20292053	-2E-12	-0.00159	6.202921	1.25E-09

P2

Step	V	F(v)	f'(v)	vi+1	delta
1	0.01	-3.75557	-599.729	0.003738	-0.00626
2	0.003738	6.452633	-4145.86	0.005294	0.001556
	0.005294	1.665234	-2100.67	0.006087	0.000793
	0.006087	0.166175	-1597.28	0.006191	0.000104
	0.006191	-0.00212	-1544.93	0.00619	-1.4E-06
	0.00619	0.000063	-1545.6	0.00619	4.08E-08
	0.00619	-1.9E-06	-1545.58	0.00619	-1.2E-09
	0.00619	5.49E-08	-1545.58	0.00619	3.55E-11

Solver	Guess	
P1	6.20285084	1.11E-07
P2	0.00618972	5.33E-08

%err1	0.00022431	PVNRT1	6.202934
%err2	0.21303689	PVNRT2	0.006203